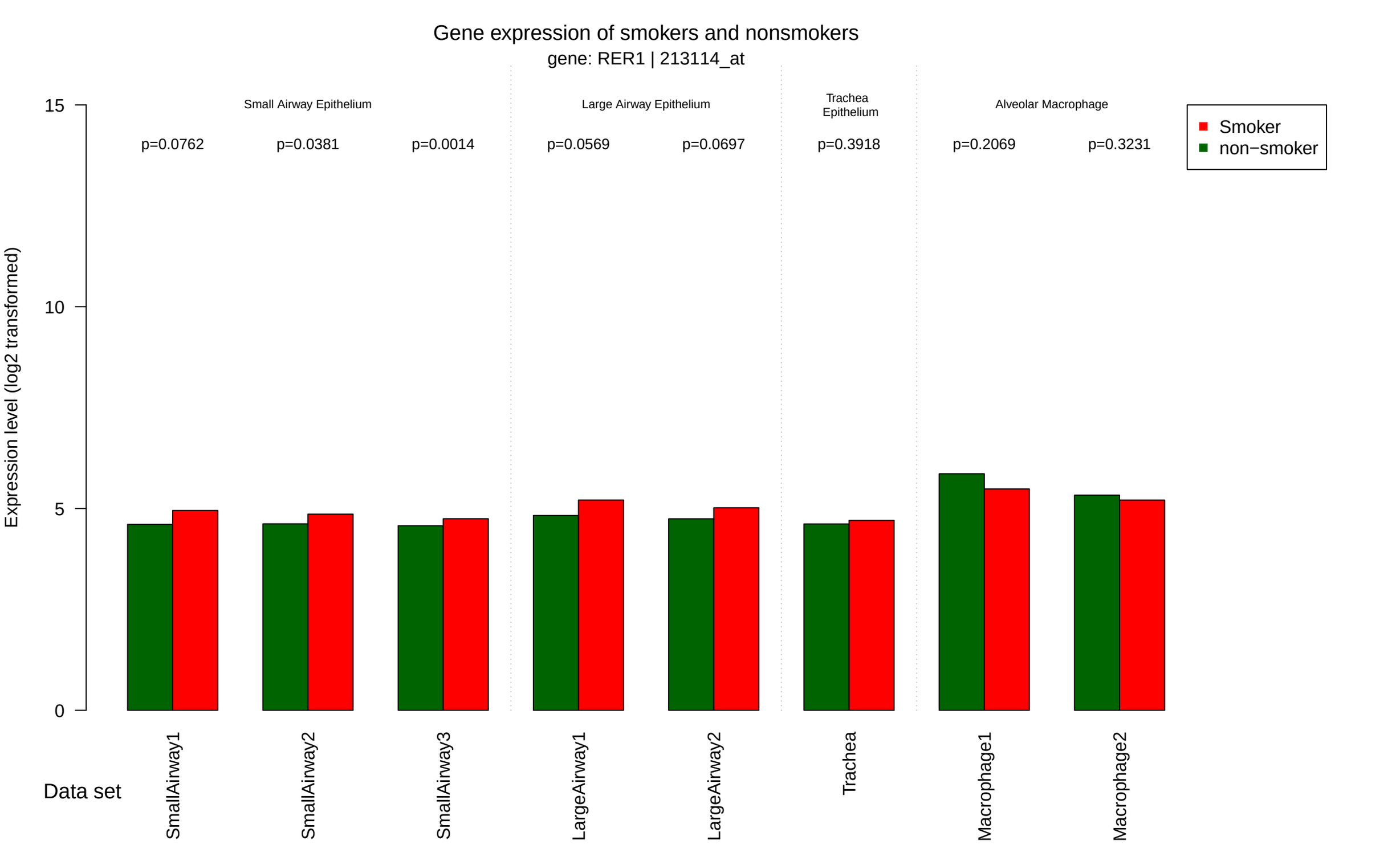
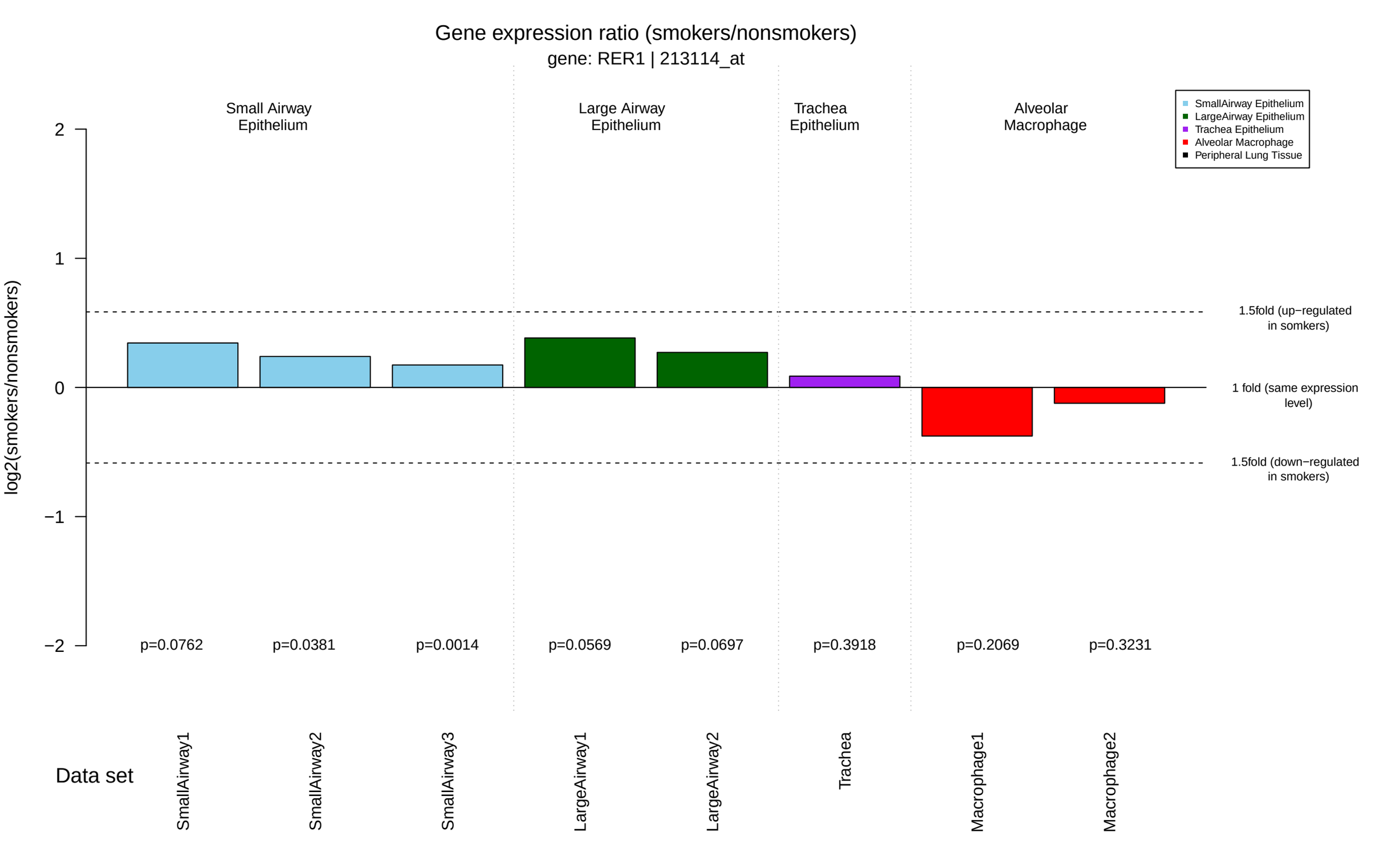
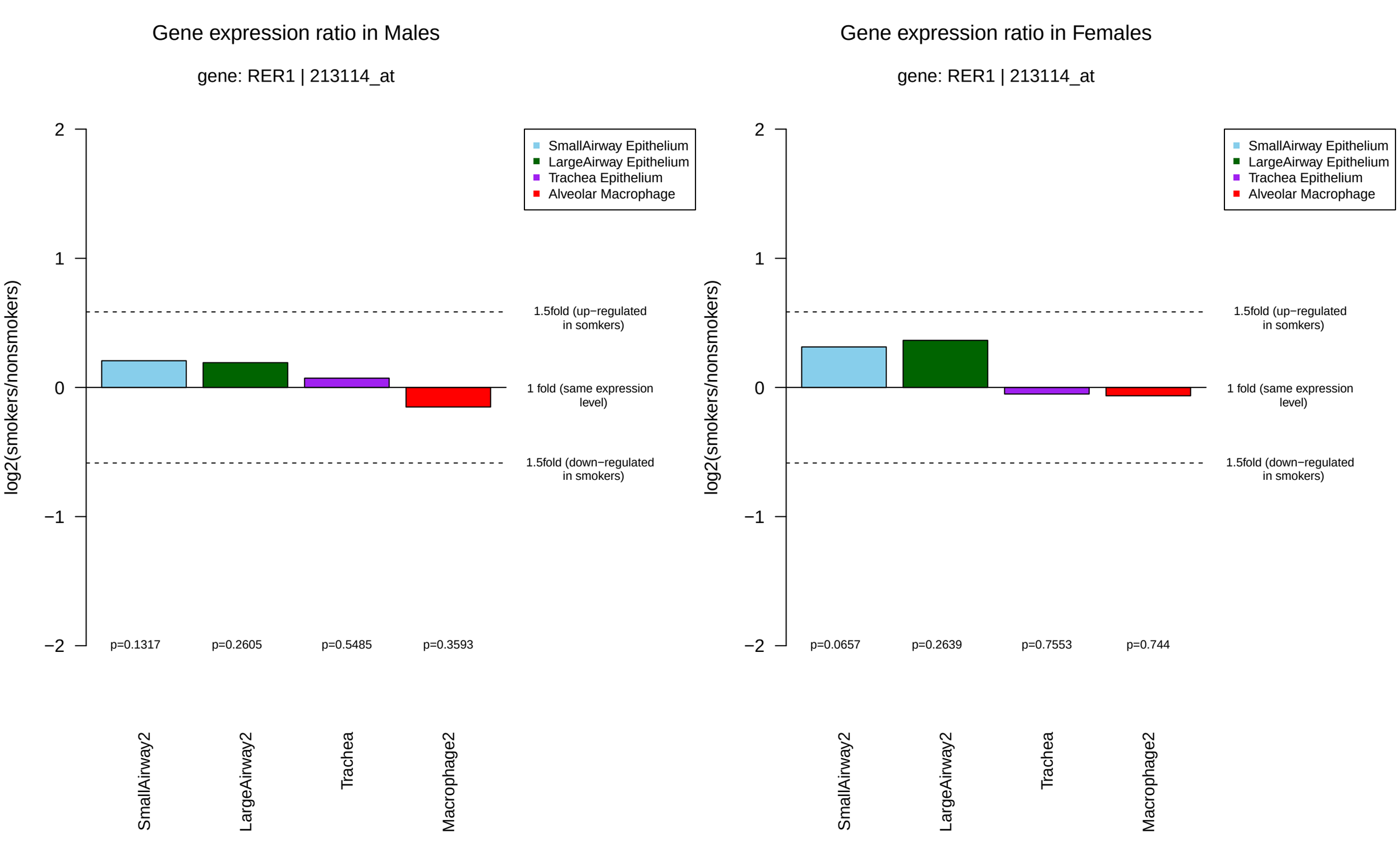
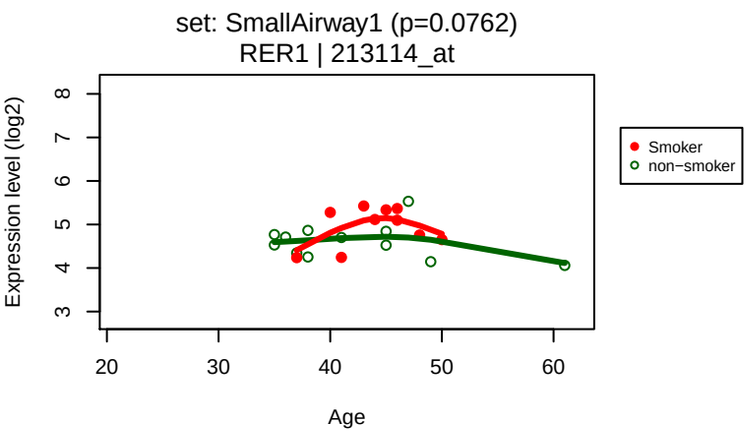
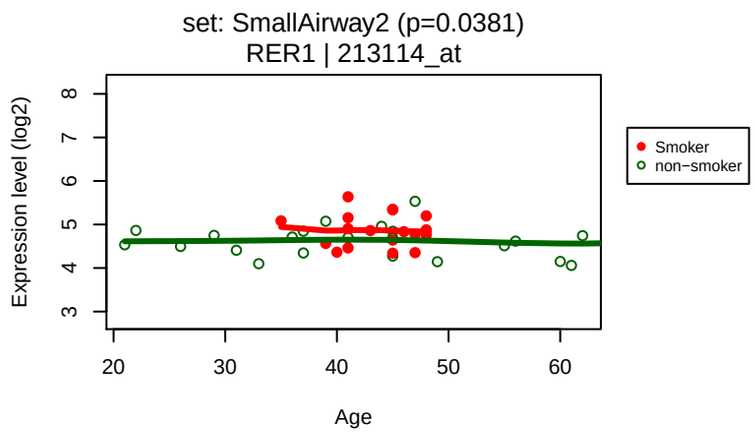
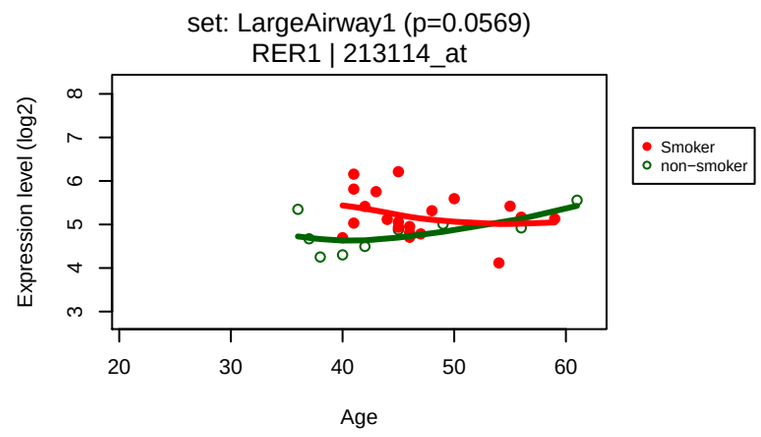

Gene expression of smokers and nonsmokers
gene: RER1 | 213114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0762
p=0.0381
p=0.0014
p=0.0569
p=0.0697
p=0.3918
p=0.2069
p=0.3231
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

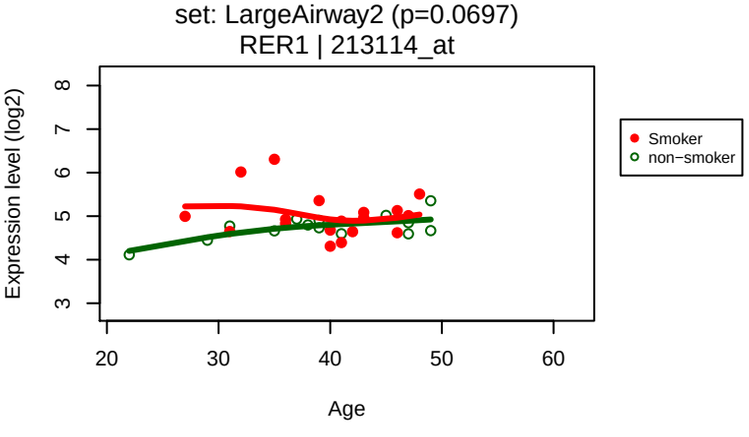

Gene expression ratio (smokers/nonsmokers)
gene: RER1 | 213114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0762
p=0.0381
p=0.0014
p=0.0569
p=0.0697
p=0.3918
p=0.2069
p=0.3231
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

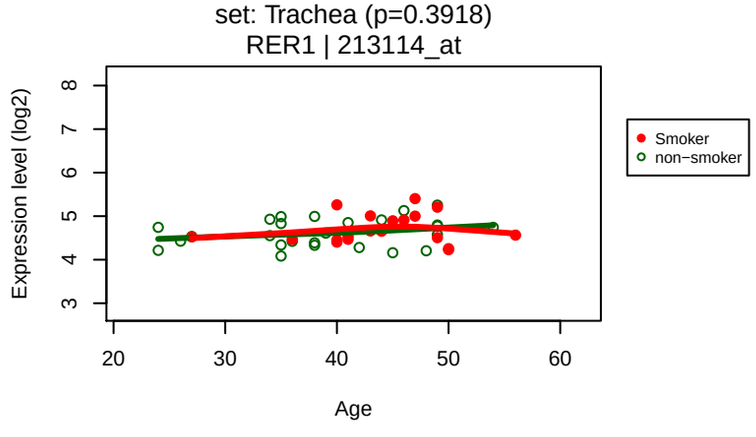

Gene expression ratio in Males
gene: RER1 | 213114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1317
p=0.2605
p=0.5485
p=0.3593
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RER1 | 213114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0657
p=0.2639
p=0.7553
p=0.744
SmallAirway2
LargeAirway2
Trachea
Macrophage2

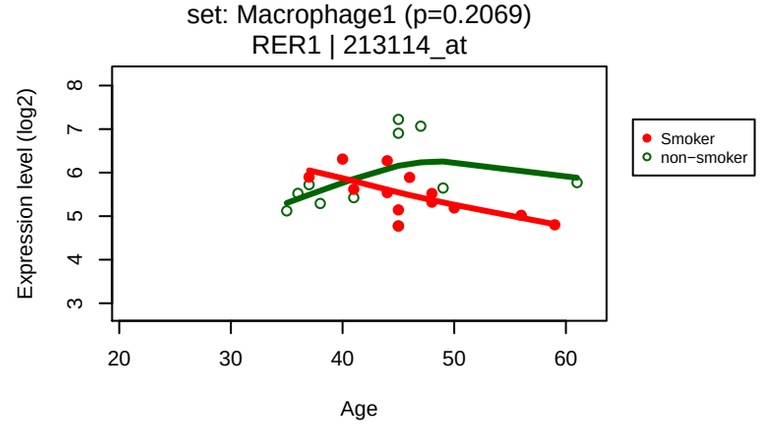

set: SmallAirway1 (p=0.0762)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker

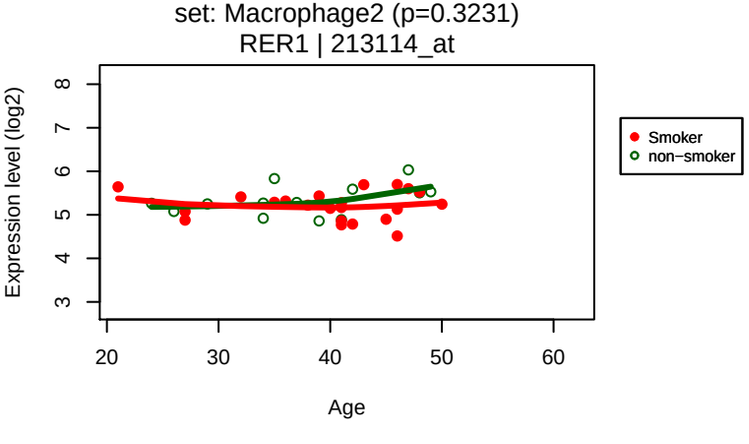

set: SmallAirway2 (p=0.0381)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker

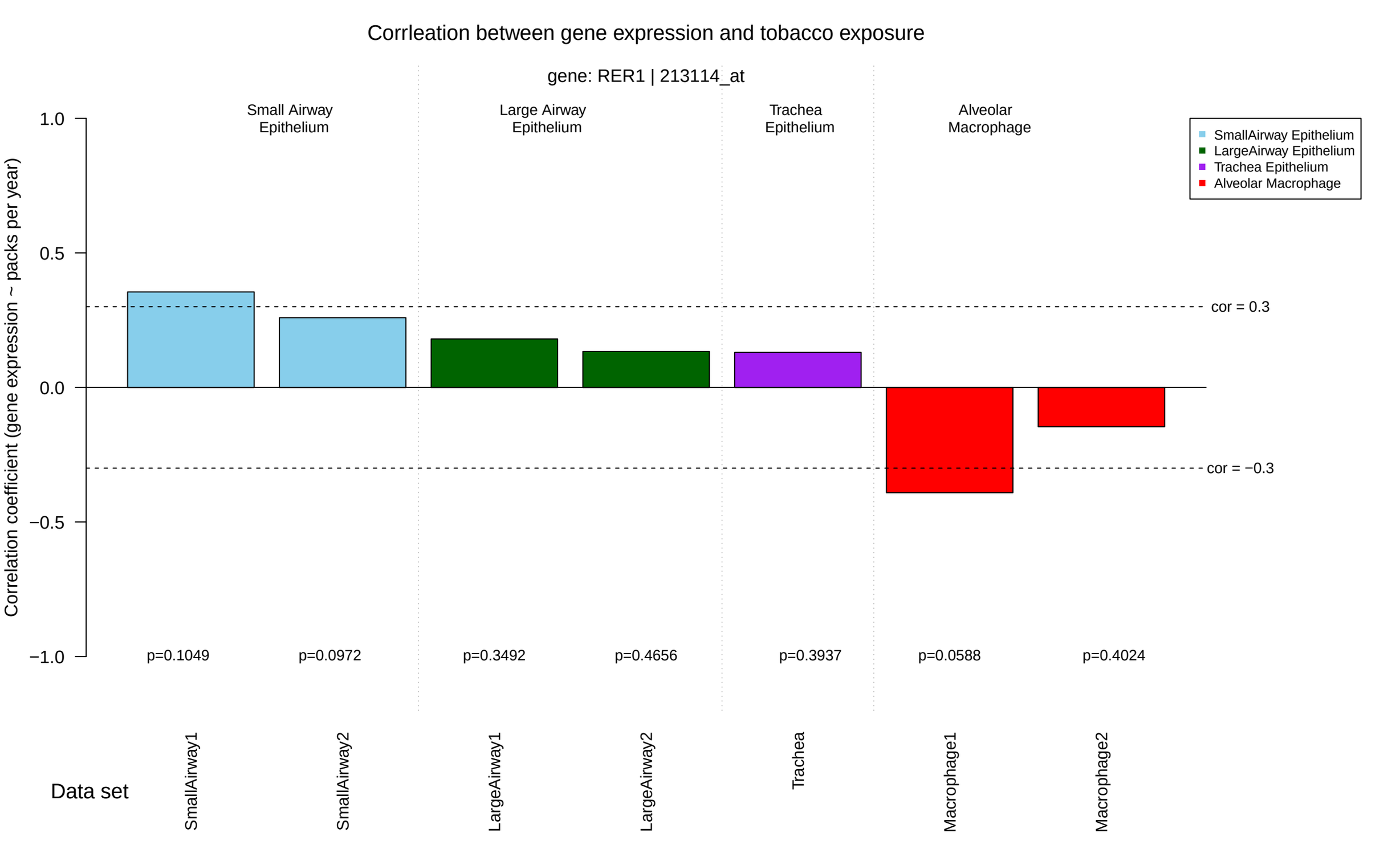

set: LargeAirway1 (p=0.0569)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0697)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3918)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2069)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3231)
RER1 | 213114_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RER1 | 213114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1049
p=0.0972
p=0.3492
p=0.4656
p=0.3937
p=0.0588
p=0.4024
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2